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ELECTRICAL DESIGN PRODUCTION

DRAFTING PORTFOLIO

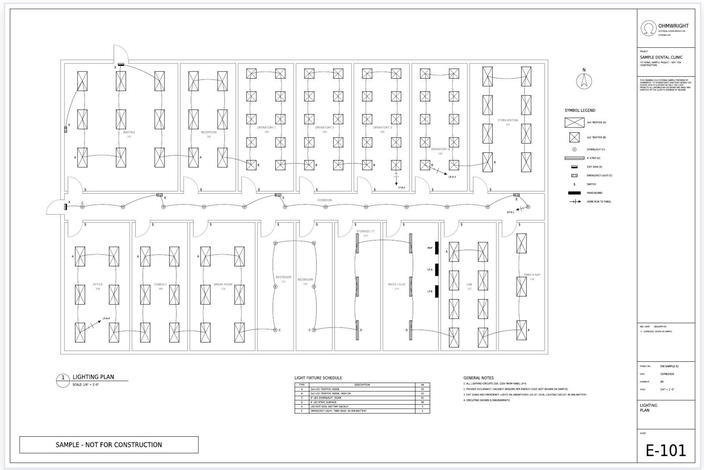
Electrical drawing sets, produced in your standards.

Power and lighting plans, one-line diagrams and panel schedules in AutoCAD and Revit, drafted from your engineer's markups and delivered in your templates.

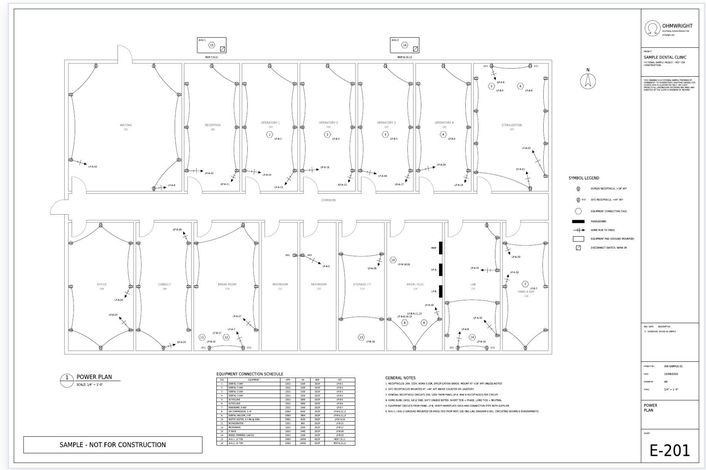
Andrew Rice
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ohmwright.com

Sample Dental Clinic

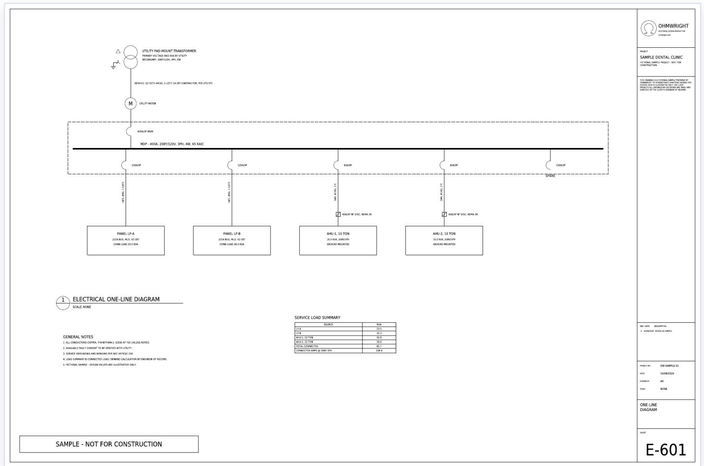
Fictional sample project · 4 sheets · ARCH D



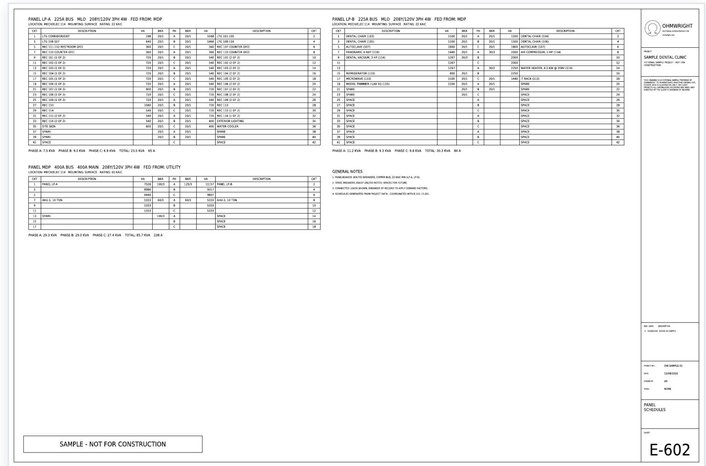
E-101 Lighting Plan
Fixture layout, switching, circuiting and fixture schedule



E-201 Power Plan
Receptacles, GFCI locations, equipment connections and schedule



E-601 One-Line Diagram
Service, distribution, feeders and connected-load summary



E-602 Panel Schedules
MDP, LP-A and LP-B with phase loading, coordinated with the plans

Coordinated

Every circuit on the plans is checked against the panel schedules before delivery.

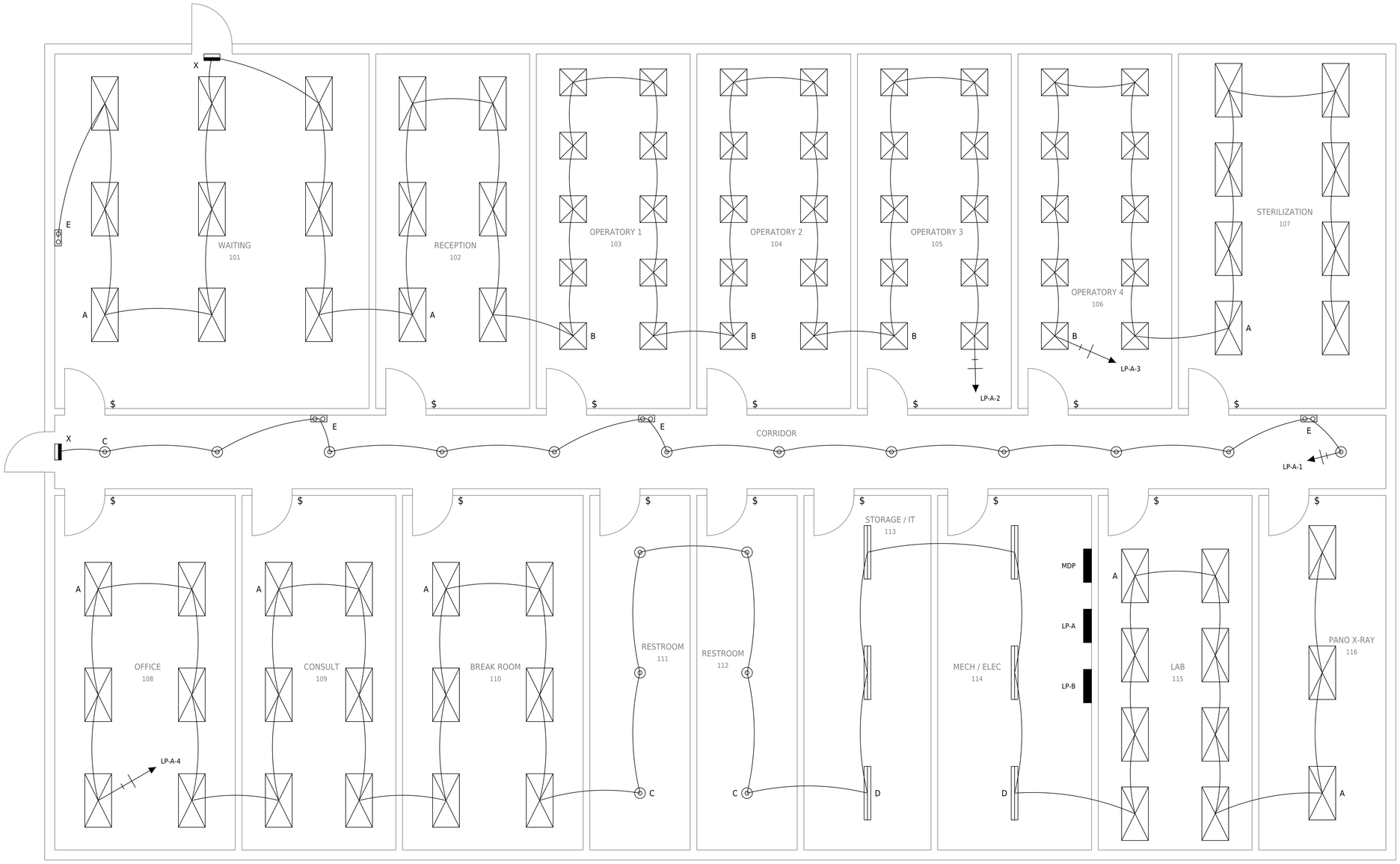
Clean sheets

A QC pass checks every sheet for overlapping text, symbols and linework.

Your format

Delivered as DWG, RVT and PDF in your title blocks, layers and Revit version.

This set is a fictional sample prepared by Ohmwright to show drafting capability. Design values are illustrative only. On client projects, all engineering decisions are made and directed by the client's engineer of record. Ohmwright does not offer professional engineering services.



1 LIGHTING PLAN

SCALE: 1/4" = 1'-0"

LIGHT FIXTURE SCHEDULE

TYPE	DESCRIPTION	VA
A	2x4 LED TROFFER, 4000K	40
B	2x2 LED TROFFER, 4000K, HIGH CRI	32
C	6" LED DOWNLIGHT, 3500K	15
D	4' LED STRIP, SURFACE	36
X	LED EXIT SIGN, BATTERY BACKUP	3
E	EMERGENCY LIGHT, TWIN HEAD, 90 MIN BATTERY	5

GENERAL NOTES

1. ALL LIGHTING CIRCUITS 20A, 120V FROM PANEL LP-A.
2. PROVIDE OCCUPANCY / VACANCY SENSORS PER ENERGY CODE (NOT SHOWN ON SAMPLE).
3. EXIT SIGNS AND EMERGENCY LIGHTS ON UNSWITCHED LEG OF LOCAL LIGHTING CIRCUIT; 90 MIN BATTERY.
4. CIRCUITING SHOWN IS DIAGRAMMATIC.

N

SYMBOL LEGEND

- 2x4 TROFFER (A)
- 2x2 TROFFER (B)
- DOWNLIGHT (C)
- 4' STRIP (D)
- EXIT SIGN (X)
- EMERGENCY LIGHT (E)
- SWITCH ($)
- PANELBOARD
- HOME RUN TO PANEL

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PROJECT

SAMPLE DENTAL CLINIC

FICTIONAL SAMPLE PROJECT - NOT FOR CONSTRUCTION

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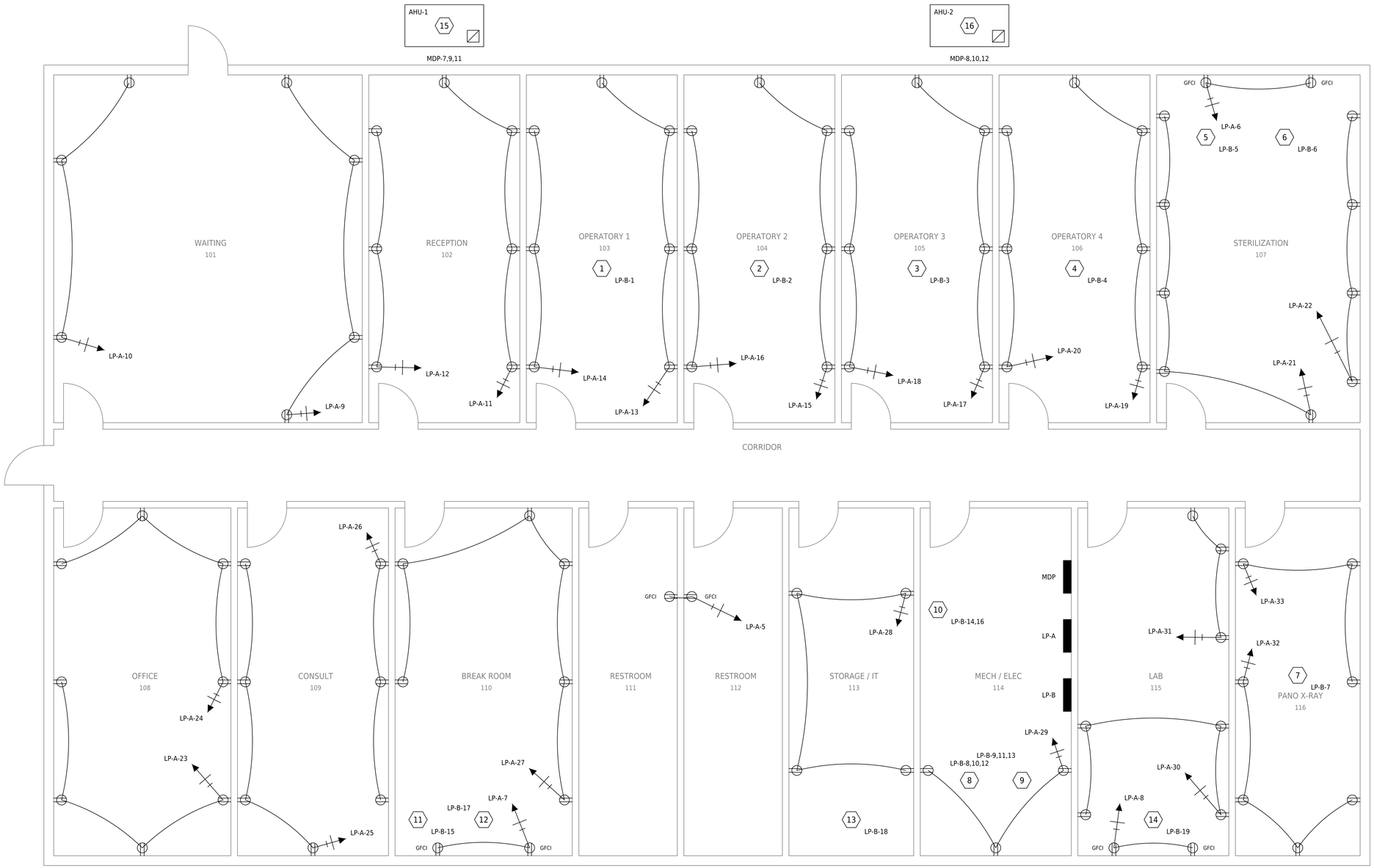
REV	DATE	DESCRIPTION
0	10/08/2026	ISSUED AS SAMPLE

PROJECT NO.	OW-SAMPLE-01
DATE	10/08/2026
DRAWN BY	AR
SCALE	1/4" = 1'-0"

LIGHTING PLAN

SHEET

E-101



SYMBOL LEGEND

- DUPLEX RECEPTACLE, +18" AFF
- GFCI RECEPTACLE, +44" AFF
- EQUIPMENT CONNECTION (TAG)
- PANELBOARD
- HOME RUN TO PANEL
- EQUIPMENT PAD (GROUND MOUNTED)
- DISCONNECT SWITCH, NEMA 3R

1 POWER PLAN

SCALE: 1/4" = 1'-0"

EQUIPMENT CONNECTION SCHEDULE

TAG	EQUIPMENT	V/PH	VA	BKR	CKT
1	DENTAL CHAIR	120/1	1500	20/1P	LP-B-1
2	DENTAL CHAIR	120/1	1500	20/1P	LP-B-2
3	DENTAL CHAIR	120/1	1500	20/1P	LP-B-3
4	DENTAL CHAIR	120/1	1500	20/1P	LP-B-4
5	AUTOCLAVE	120/1	1800	20/1P	LP-B-5
6	AUTOCLAVE	120/1	1800	20/1P	LP-B-6
7	PANORAMIC X-RAY	120/1	1440	20/1P	LP-B-7
8	AIR COMPRESSOR, 5 HP	208/3	6000	30/3P	LP-B-8,10,12
9	DENTAL VACUUM, 3 HP	208/3	3800	20/3P	LP-B-9,11,13
10	WATER HEATER, 4.5 KW @ 208V	208/1	4500	30/2P	LP-B-14,16
11	REFRIGERATOR	120/1	800	20/1P	LP-B-15
12	MICROWAVE	120/1	1500	20/1P	LP-B-17
13	IT RACK	120/1	1440	20/1P	LP-B-18
14	MODEL TRIMMER / LAB EQ	120/1	1200	20/1P	LP-B-19
15	AHU-1, 10 TON	208/3	16000	60/3P	MDP-7,9,11
16	AHU-2, 10 TON	208/3	16000	60/3P	MDP-8,10,12

GENERAL NOTES

1. RECEPTACLES: 20A, 125V, NEMA 5-20R, SPECIFICATION GRADE. MOUNT AT +18" AFF UNLESS NOTED.
2. GFCI RECEPTACLES MOUNTED AT +44" AFF ABOVE COUNTER OR LAVATORY.
3. GENERAL RECEPTACLE CIRCUITS 20A, 120V FROM PANEL LP-A; MAX 6 RECEPTACLES PER CIRCUIT.
4. HOME RUNS: 2#12, 1#12 GND, 3/4"C UNLESS NOTED. SHORT TICK = PHASE, LONG TICK = NEUTRAL.
5. EQUIPMENT CIRCUITS FROM PANEL LP-B; VERIFY NAMEPLATE DATA AND CONNECTION TYPE WITH SUPPLIER.
6. AHU-1 / AHU-2 (GROUND MOUNTED ON PADS) FED FROM MDP; SEE ONE-LINE DIAGRAM E-601. CIRCUITING SHOWN IS DIAGRAMMATIC.

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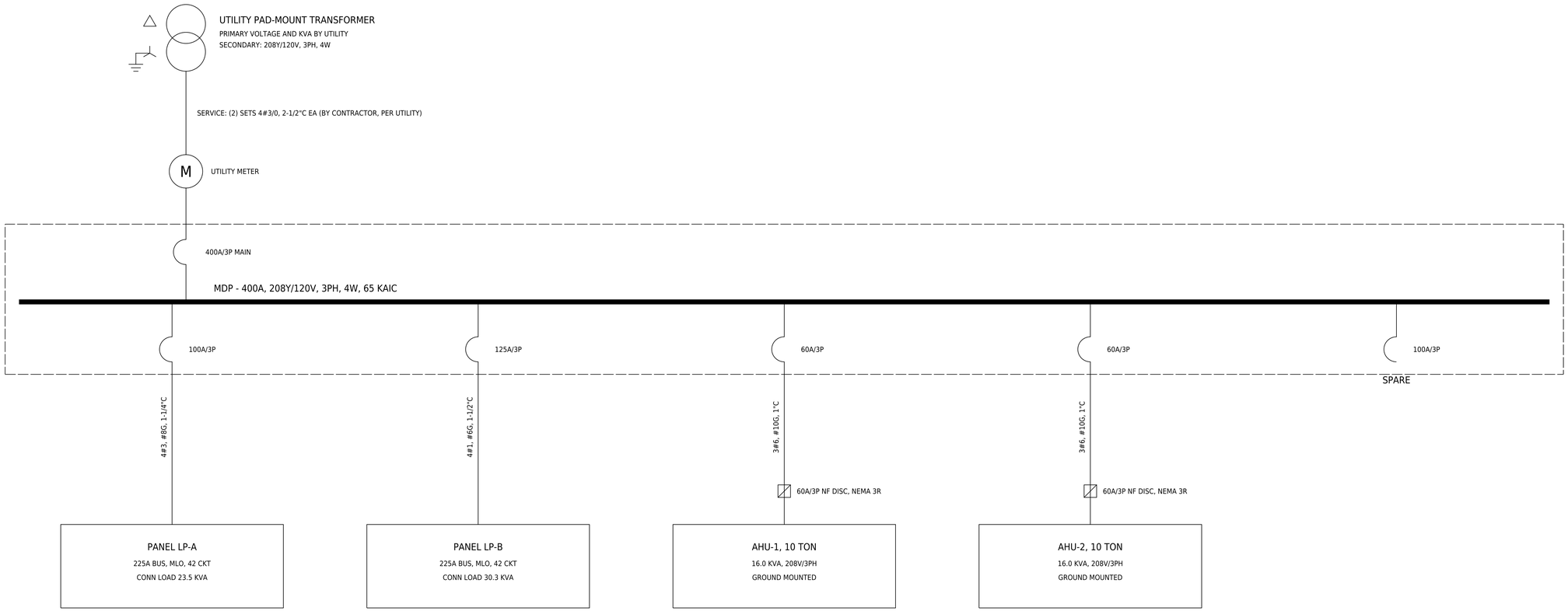
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POWER PLAN

SHEET

E-201



1 ELECTRICAL ONE-LINE DIAGRAM

SCALE: NONE

SERVICE LOAD SUMMARY

SOURCE	KVA
LP-A	23.5
LP-B	30.3
AHU-1, 10 TON	16.0
AHU-2, 10 TON	16.0
TOTAL CONNECTED	85.7
CONNECTED AMPS @ 208V 3PH	238 A

GENERAL NOTES

1. ALL CONDUCTORS COPPER, THHN/THWN-2, SIZED AT 75C UNLESS NOTED.
2. AVAILABLE FAULT CURRENT TO BE VERIFIED WITH UTILITY.
3. SERVICE GROUNDING AND BONDING PER NEC ARTICLE 250.
4. LOAD SUMMARY IS CONNECTED LOAD; DEMAND CALCULATION BY ENGINEER OF RECORD.
5. FICTIONAL SAMPLE - DESIGN VALUES ARE ILLUSTRATIVE ONLY.

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ONE-LINE DIAGRAM

SHEET

E-601

PANEL LP-A 225A BUS MLO 208Y/120V 3PH 4W FED FROM: MDP

LOCATION: MECH/ELEC 114 MOUNTING: SURFACE RATING: 22 KAIC

CKT	DESCRIPTION	VA	BKR	PH	BKR	VA	DESCRIPTION	CKT
1	LTG CORRIDOR/EXIT	198	20/1	A	20/1	1568	LTG 101-105	2
3	LTG 106-107	640	20/1	B	20/1	1466	LTG 108-116	4
5	REC 111-112 RESTROOM GFCI	360	20/1	C	20/1	360	REC 107 COUNTER GFCI	6
7	REC 110 COUNTER GFCI	360	20/1	A	20/1	360	REC 115 COUNTER GFCI	8
9	REC 101 (1 OF 2)	720	20/1	B	20/1	540	REC 101 (2 OF 2)	10
11	REC 102 (1 OF 2)	720	20/1	C	20/1	540	REC 102 (2 OF 2)	12
13	REC 103 (1 OF 2)	720	20/1	A	20/1	540	REC 103 (2 OF 2)	14
15	REC 104 (1 OF 2)	720	20/1	B	20/1	540	REC 104 (2 OF 2)	16
17	REC 105 (1 OF 2)	720	20/1	C	20/1	540	REC 105 (2 OF 2)	18
19	REC 106 (1 OF 2)	720	20/1	A	20/1	540	REC 106 (2 OF 2)	20
21	REC 107 (1 OF 2)	900	20/1	B	20/1	720	REC 107 (2 OF 2)	22
23	REC 108 (1 OF 2)	720	20/1	C	20/1	720	REC 108 (2 OF 2)	24
25	REC 109 (1 OF 2)	720	20/1	A	20/1	540	REC 109 (2 OF 2)	26
27	REC 110	1080	20/1	B	20/1	720	REC 113	28
29	REC 114	540	20/1	C	20/1	720	REC 115 (1 OF 2)	30
31	REC 115 (2 OF 2)	540	20/1	A	20/1	720	REC 116 (1 OF 2)	32
33	REC 116 (2 OF 2)	540	20/1	B	20/1	400	EXTERIOR LIGHTING	34
35	SITE SIGN	600	20/1	C	20/1	400	WATER COOLER	36
37	SPARE		20/1	A	20/1		SPARE	38
39	SPARE		20/1	B	20/1		SPARE	40
41	SPACE			C			SPACE	42

PHASE A: 7.5 KVA PHASE B: 9.0 KVA PHASE C: 6.9 KVA TOTAL: 23.5 KVA 65 A

PANEL MDP 400A BUS 400A MAIN 208Y/120V 3PH 4W FED FROM: UTILITY

LOCATION: MECH/ELEC 114 MOUNTING: SURFACE RATING: 65 KAIC

CKT	DESCRIPTION	VA	BKR	PH	BKR	VA	DESCRIPTION	CKT
1	PANEL LP-A	7526	100/3	A	125/3	11157	PANEL LP-B	2
3		8986		B		9317		4
5		6940		C		9807		6
7	AHU-1, 10 TON	5333	60/3	A	60/3	5333	AHU-2, 10 TON	8
9		5333		B		5333		10
11		5333		C		5333		12
13	SPARE		100/3	A			SPACE	14
15				B			SPACE	16
17				C			SPACE	18

PHASE A: 29.3 KVA PHASE B: 29.0 KVA PHASE C: 27.4 KVA TOTAL: 85.7 KVA 238 A

PANEL LP-B 225A BUS MLO 208Y/120V 3PH 4W FED FROM: MDP

LOCATION: MECH/ELEC 114 MOUNTING: SURFACE RATING: 22 KAIC

CKT	DESCRIPTION	VA	BKR	PH	BKR	VA	DESCRIPTION	CKT
1	DENTAL CHAIR (103)	1500	20/1	A	20/1	1500	DENTAL CHAIR (104)	2
3	DENTAL CHAIR (105)	1500	20/1	B	20/1	1500	DENTAL CHAIR (106)	4
5	AUTOCLAVE (107)	1800	20/1	C	20/1	1800	AUTOCLAVE (107)	6
7	PANORAMIC X-RAY (116)	1440	20/1	A	30/3	2000	AIR COMPRESSOR, 5 HP (114)	8
9	DENTAL VACUUM, 3 HP (114)	1267	20/3	B		2000		10
11		1267		C		2000		12
13		1267		A	30/2	2250	WATER HEATER, 4.5 KW @ 208V (114)	14
15	REFRIGERATOR (110)	800	20/1	B		2250		16
17	MICROWAVE (110)	1500	20/1	C	20/1	1440	IT RACK (113)	18
19	MODEL TRIMMER / LAB EQ (115)	1200	20/1	A	20/1		SPARE	20
21	SPARE		20/1	B	20/1		SPARE	22
23	SPARE		20/1	C			SPACE	24
25	SPACE			A			SPACE	26
27	SPACE			B			SPACE	28
29	SPACE			C			SPACE	30
31	SPACE			A			SPACE	32
33	SPACE			B			SPACE	34
35	SPACE			C			SPACE	36
37	SPACE			A			SPACE	38
39	SPACE			B			SPACE	40
41	SPACE			C			SPACE	42

PHASE A: 11.2 KVA PHASE B: 9.3 KVA PHASE C: 9.8 KVA TOTAL: 30.3 KVA 84 A

GENERAL NOTES

1. PANELBOARDS: BOLTED BREAKERS, COPPER BUS, 22 KAIC MIN (LP-A, LP-B).
2. SPARE BREAKERS 20A/1P UNLESS NOTED; SPACES FOR FUTURE.
3. CONNECTED LOADS SHOWN; ENGINEER OF RECORD TO APPLY DEMAND FACTORS.
4. SCHEDULES GENERATED FROM PROJECT DATA - COORDINATED WITH E-101 / E-201.

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SCALE	NONE

PANEL SCHEDULES

SHEET

E-602